

Fully Autonomous Calibration for Astrophysics via Adaptive Parameter-Space Exploration

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ABSTRACT

Astrophysical simulations and semi-analytic models contain parameters that cannot always be fixed from first principles, making calibration expensive when each model run is costly. We present a fully autonomous calibration framework that uses normalizing flows to adaptively explore parameter space and select promising new model evaluations. The workflow is designed to be reusable across different astrophysical models and targets. We test the method on three problems: Sedov–Taylor blast waves, Rotating Cylinder angular-momentum conservation, and SAGE galaxy formation, covering both AMR parameters and semi-analytic feedback parameters. In SAGE, the method improves the combined fit to eight observational constraints: the best single-flow direct branch increases R^2 from 0.4359 for the shared best initial design to 0.7060 after 10 active evaluations, while a five-member ensemble-MAP branch reaches $R^2 = 0.6851$ with a seed-aware acquisition rule. The Sedov–Taylor tests also show that the method can improve calibration accuracy while accounting for simulation cost. These results show that adaptive parameter-space exploration can provide a simple, transferable, and computationally efficient approach to autonomous calibration in astrophysics.

Keywords: Astronomical simulations (1857) — Hydrodynamical simulations (767) — Semi-analytic models (1476) — Computational methods (1965) — Astronomy data modeling (1859)

1. INTRODUCTION

Galaxy formation theory provides the physical frame for this work (R. S. Somerville & R. Davé 2015). In the standard picture, galaxies form and evolve within a hierarchy of dark-matter halos, while gas accretion, radiative cooling, star formation, stellar feedback, black-hole growth, active galactic nucleus feedback, and chemical enrichment regulate the baryonic content of those halos. The quantities used to test this theory are observable summaries of the coupled system, including stellar-mass functions, star-formation histories, gas fractions, metallicities, and correlations between black holes and their host bulges (I. K. Baldry et al. 2012; P. Madau & M. Dickinson 2014; B. Catinella et al. 2018; A. Gallazzi et al. 2005; J. Kormendy & L. C. Ho 2013).

Numerical models turn this theory into predictions (V. Springel et al. 2005; M. Vogelsberger et al. 2020). Hydrodynamical simulations solve the fluid and gravitational evolution directly while introducing numerical choices such as resolution, mesh refinement, and subgrid closures (J. Schaye et al. 2015; R. Davé et al. 2019); semi-analytic models evolve galaxies on dark-matter halo merger trees through parameterized prescriptions for cooling, star formation, feedback, and en-

vironmental processing (G. Kauffmann et al. 1993; D. J. Croton et al. 2016). In both settings, the physical prediction depends on adjustable parameters and algorithmic choices that cannot be fixed from first principles alone. The same issue appears in controlled hydrodynamic tests, where the target may be a shock profile or a conservation history, and in galaxy-formation models, where the target may be a vector of observational summary statistics.

Calibration is therefore a central modeling problem. Given a simulator, a target representation, and a limited evaluation budget, the task is to identify parameter combinations that improve agreement with the physical or observational target while keeping the link between the target and the underlying astrophysics explicit.

The calibration problem is difficult because the target surface is structured, correlated, and expensive to sample. A broad parameter sweep can spend most of its budget in regions that are already inconsistent with the target, while a single scalar acquisition score can hide physically important trade-offs among profiles or observable families. Statistical calibration is often framed as inference over model parameters, observational data, and model discrepancy (M. C. Kennedy & A. O’Hagan 2001). When new evaluations must be chosen sequen-

tially, the same problem also connects to active learning and Bayesian optimization (D. R. Jones et al. 1998; B. Shahriari et al. 2016; P. I. Frazier 2018).

Existing machine-learning approaches have made this search more practical. ML-based calibration and active learning can learn from completed simulations, predict promising regions of parameter space, and select new evaluations more efficiently than unguided sweeps. Implicit-likelihood inference has also been used to calibrate cosmological simulations against galaxy growth observables (Y. Jo et al. 2023). Machine-guided Enzo calibration has combined neural density estimation with active learning to recommend subgrid-parameter simulations (B. K. Oh et al. 2022). Cosmological hydrodynamical programs such as FLAMINGO use Latin-hypercube designs, Gaussian-process emulators, and Bayesian fitting to calibrate feedback parameters against galaxy and cluster observables (R. Kugel et al. 2023). Semi-analytic model calibration has been accelerated with Particle Swarm Optimization, deep-learning emulators, and reduced-cost Galacticus likelihoods (A. N. Ruiz et al. 2013; E. J. Elliott et al. 2021; A. Robertson & A. Benson 2025).

Normalizing flows provide an attractive density-estimation tool in this broader landscape because they combine flexible nonlinear transformations with tractable density evaluation and efficient sampling. Recent cosmology and physical-science applications use flows to compress, deploy, and draw samples from complex posterior distributions (Y. Yamauchi et al. 2023); to learn probe-specific parameter distributions that can be combined while marginalizing nuisance parameters and reducing repeated expensive likelihood evaluations (A. Mootoovaloo et al. 2024); and to approximate simulation-driven, selection-aware likelihoods that are difficult to write analytically but can be reused across cosmological models (B. M. Boyd et al. 2026). In parallel, simulation-based inference, including sequential neural posterior estimation and automatic posterior transformation, provides frameworks for learning posterior distributions when the likelihood is unavailable, with software such as LtU-ILI emphasizing posterior validation and coverage diagnostics across astrophysical and cosmological applications (D. S. Greenberg et al. 2019; K. Cranmer et al. 2020; A. Tejero-Cantero et al. 2020; M. Ho et al. 2024).

In this work, we use the same normalizing-flow machinery for a different purpose. Unlike these posterior-inference workflows, the primary object of interest here is the best realized calibration result under a small number of completed model evaluations, not a fully calibrated Bayesian posterior over all physical parameters.

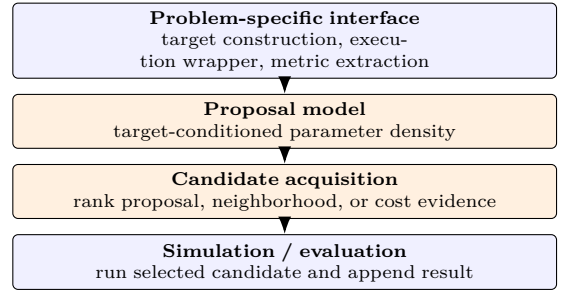


Figure 1. Reusable abstraction for the sequential calibration workflow. The case-specific interface defines the physical target and model wrapper, while the proposal and acquisition steps are reused across the Sedov–Taylor, Rotating Cylinder, and SAGE cases.

Because the training set is adaptively expanded along the acquisition trajectory, the learned parameter density can depend on the proposal path. We therefore assess the success of the calibration procedure primarily through best-so-far target agreement as a function of simulation budget, while treating corner-plot stability as a secondary diagnostic rather than the main convergence criterion.

We call this strategy normalizing-flow-guided sequential active calibration: a workflow in which a learned inverse proposal model suggests where the simulation or astrophysical model is evaluated next, while the physical target and acquisition rule remain explicit. Figure 1 summarizes the reusable abstraction. The learned proposal is implemented with a conditional rational-quadratic spline flow (CRQSF), a specific conditional normalizing-flow architecture, (G. Papamakarios et al. 2017; C. Durkan et al. 2019; G. Papamakarios et al. 2021).

The evidence is staged to test calibration accuracy, target representation, and portability rather than to optimize one simulator. Sedov–Taylor is the controlled hydrodynamic benchmark for target construction, profile context, small seed sets, and optional cost signals. Rotating Cylinder is a second Enzo closed-loop test: it keeps the profile-conditioned acquisition interface but changes the physical target to angular-momentum conservation and exercises a runtime-aware branch. Local SAGE is the astrophysical transfer example. It uses the same outer orchestration on a semi-analytic galaxy-formation model, calibrating SAGE-paper scale parameters against multi-family observational targets.

The structure of this paper is as follows. In Section 2, we define the reusable calibration framework, including the initial training set, CRQSF proposal model, candidate acquisition rule, cost-aware extension, and the Sedov–Taylor, Rotating Cylinder, and SAGE case-

specific implementations. In Section 3, we present the Sedov–Taylor, Rotating Cylinder, and SAGE calibration results, including cost-aware AMR behavior and learned SAGE parameter structure. In Section 4, we discuss AMR resource management, SAGE interpretation, transferability, and limitations. In Section 5, we summarize the main results and future work, with extended Sedov parameter-sensitivity diagnostics given in the Appendix.

2. METHOD

2.1. Calibration Framework

To separate the reusable engine from the case-specific interfaces, let $\theta \in \Theta$ denote the calibrated parameter vector and let $y(\theta)$ denote the simulation or model output after case-specific post-processing. The target representation is $c_\star$, which may be an analytical profile, a derived physical diagnostic, or an observational metric vector. Each evaluated row is summarized by

$$\mathcal{D}_n = \{(\theta_i, c_i, E_i, C_i)\}_{i=1}^n, \quad (1)$$

where c_i is the extracted context, E_i is the calibration error or loss, and C_i is an optional cost measurement. The primary objective is calibration optimization,

$$\theta^\star = \arg \min_{\theta \in \Theta} J(\theta), \quad (2)$$

where J denotes the case-specific mismatch between the simulated output and the target. At any finite budget, the recommended parameter is the best observed row,

$$\begin{aligned} \theta_{\text{best}}(n) &= \arg \min_{\theta_i \in \mathcal{D}_n} J(\theta_i), \\ J_{\text{best}}(n) &= \min_{\theta_i \in \mathcal{D}_n} J(\theta_i). \end{aligned} \quad (3)$$

Thus the active-learning objective is not to estimate a final Bayesian posterior in one step. Instead, the engine learns a target-conditioned calibration density over useful parameter values and uses that density to propose new evaluations. Convergence is assessed through J_{best} or its R^2 -equivalent as a function of the number of completed simulations, not through visual convergence of the learned density alone.

The separation between the problem-specific interface and the reusable engine is central to the framework. The interface is responsible for target construction, execution, and metric extraction: it converts the scientific goal into a conditioning context, turns a proposed parameter vector into a completed model evaluation, and reduces the completed output to the quantities stored in the calibration table.

The automated engine has three reusable components: a CRQSF target-conditioned proposal density, finite-pool candidate acquisition, and, for cost-aware branches,

an optional cost surrogate. The Sedov cost-aware experiment uses an automatic-relevance Gaussian-process regressor for refinement-complexity prediction. Across Sedov, Rotating Cylinder, and SAGE, only the target construction, execution wrapper, and metric extractor are case-specific.

The sequential calibration workflow is an active-learning loop over completed evaluations. Starting from evaluated seed rows, iteration r normalizes the current parameters and contexts, retrain the inverse proposal density on $\mathcal{D}_r$, conditions the proposal on $c_\star$, draws and scores a finite candidate pool, evaluates only the selected candidate or small batch, and appends the new row to form $\mathcal{D}_{r+1}$. This train–propose–acquire–evaluate cycle repeats until the evaluation budget or tolerance criterion is reached.

The evaluation protocol reports calibration accuracy and, when available, computational cost. Accuracy is measured by the case-specific target mismatch or its equivalent best-so-far agreement score under the completed evaluation budget. Computational cost is reported when a reliable cost signal is available and is interpreted through Pareto or marginal-efficiency diagnostics when cost-aware acquisition is used.

2.2. Initial Training Set

Astrophysical calibration usually begins before any reliable local surrogate is available. The initial design must therefore cover the allowed physical parameter volume without committing to a single optimum too early. In closed-loop simulation cases, Latin-hypercube sampling (M. D. McKay et al. 1979) is used to create the seed table before the first learned proposal model is trained. For an initial design with n_0 rows and d_θ parameters, the unit-cube coordinate is

$$\begin{aligned} u_{ij} &\sim \mathcal{U}((\pi_j(i) - 1)/n_0, \pi_j(i)/n_0), \\ i &= 1, \dots, n_0, \quad j = 1, \dots, d_\theta. \end{aligned} \quad (4)$$

The physical parameter transform is case-specific:

$$\begin{aligned} \theta_{ij} &= a_j + (b_j - a_j)u_{ij} && \text{(linear),} \\ \theta_{ij} &= 10^{\log_{10} a_j + (\log_{10} b_j - \log_{10} a_j)u_{ij}} && \text{(log).} \end{aligned} \quad (5)$$

After these seed rows are evaluated or reduced, they become the initial completed table used to train the learned proposal model.

2.3. Conditional Rational-Quadratic Spline Flow Proposal

After seed rows are evaluated, the calibration problem becomes inverse: the target is known, but the parameter vectors likely to reproduce it are not. Let θ denote a

simulation parameter vector, and let c denote the context extracted from a completed simulation or reduction. We use “normalizing flow” to denote the general class of invertible density models, and CRQSF to denote the specific conditional spline architecture used for the proposal model in this paper. CRQSF learns

$$q_\phi(\boldsymbol{\theta} \mid c). \quad (6)$$

If the completed rows were independent simulations with $\boldsymbol{\theta} \sim p_{\text{prior}}(\boldsymbol{\theta})$, then a sufficiently flexible conditional density estimator trained by maximum likelihood would approximate the prior-defined inverse density

$$\pi_0(\boldsymbol{\theta} \mid c_\star) \propto p(c_\star \mid \boldsymbol{\theta}) p_{\text{prior}}(\boldsymbol{\theta}), \quad (7)$$

which is the neural-posterior-estimation setting used in simulation-based inference (D. S. Greenberg et al. 2019; K. Cranmer et al. 2020; A. Tejero-Cantero et al. 2020). In our sequential calibration loop, however, the table after the seed stage also contains rows selected by the current learned proposal, finite candidate-pool ranking, and, in some branches, cost-aware filtering. The marginal design density of completed parameters is therefore an adaptive distribution $m_r(\boldsymbol{\theta})$, not the original prior p_{prior} . In this calibration setting, p_{prior} should often be read as an initial design measure over the allowed parameter domain, rather than as a strong subjective astrophysical prior. This distinction is the proposal-bias issue discussed in sequential neural posterior estimation: without a proposal correction, a conditional estimator trained on adaptive simulations learns the conditional density induced by the proposal distribution,

$$\pi_{m_r}(\boldsymbol{\theta} \mid c_\star) \propto p(c_\star \mid \boldsymbol{\theta}) m_r(\boldsymbol{\theta}), \quad (8)$$

rather than the prior-defined posterior in Equation (7) (D. S. Greenberg et al. 2019; G. Papamakarios et al. 2019). Recent physical-science normalizing-flow applications avoid this ambiguity by training flows on posterior samples, using flows as reusable posterior factors, or learning a likelihood term that is later embedded in a Bayesian sampler (Y. Yamauchi et al. 2023; A. Mootooyaloo et al. 2024; B. M. Boyd et al. 2026). In the implemented positive-parameter branches, the flow is trained on the standardized log-parameter vector $\mathbf{y} = T_\theta(\boldsymbol{\theta})$, where T_θ first applies $\log_{10}$ to each positive parameter and then standardizes the log coordinates.

The flow density follows the standard normalizing-flow change-of-variables construction (D. J. Rezende & S. Mohamed 2015; G. Papamakarios et al. 2017, 2021). Let $\mathbf{z}$ be a latent vector with standard normal density $p_0(\mathbf{z})$. The flow applies a context-conditioned invertible transformation $f_\phi(\cdot; c)$ from standardized parameters $\mathbf{y}$

to $\mathbf{z}$:

$$\mathbf{z} = f_\phi(\mathbf{y}; c), \quad (9)$$

and the conditional log density is

$$\log q_\phi(\mathbf{y} \mid c) = \log p_0(f_\phi(\mathbf{y}; c)) + \log \left| \det \frac{\partial f_\phi(\mathbf{y}; c)}{\partial \mathbf{y}} \right|. \quad (10)$$

Training minimizes the negative log density over completed rows,

$$\mathcal{L}_{\text{NLL}}(\phi) = -\frac{1}{N} \sum_{i=1}^N \log q_\phi(\mathbf{y}_i \mid c_i). \quad (11)$$

For active-learning selection we use q_{ϕ_r} directly as the learned proposal. For posterior-style diagnostic plots, the density is evaluated in physical parameter space and corrected for the prior-versus-design density when the proposal design can be reconstructed:

$$\begin{aligned} \log \tilde{\pi}_r(\boldsymbol{\theta} \mid c_\star) &= \log q_{\phi_r}(T_\theta(\boldsymbol{\theta}) \mid c_\star) + \log \left| \det \frac{\partial T_\theta(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right| \\ &\quad + \log p_{\text{prior}}(\boldsymbol{\theta}) - \log \hat{m}_r(\boldsymbol{\theta}) + \text{const.} \end{aligned} \quad (12)$$

Here $\hat{m}_r$ is an estimated design density for the completed training rows: LHS rows contribute the prior design, while active rows contribute the learned proposal and any reconstructable selection rule. The deterministic Top- k step used in some branches makes the exact selection density difficult to write in closed form, so corrected corner plots are used as proposal-bias-aware diagnostics rather than as fully calibrated Bayesian posteriors. The raw context can be a fixed-grid profile tensor or an observation-facing metric vector; a context encoder maps this payload into the flow-conditioning space, so the model input does not require manually engineered physical summary features.

Operationally, the SAGE corner plot uses this correction only after the active learning run is complete. The plotted samples are not used to choose the next simulation. We first sample the final learned density in standardized log-parameter coordinates, map each retained Markov-chain state to physical parameters, and reject states outside the adopted prior bounds. For an in-bounds sample, the plotted log density is

$$\begin{aligned} \ell_c(\boldsymbol{\theta}) &= \log q_{\phi_r}(\mathbf{y} \mid c_\star) + J_T(\boldsymbol{\theta}) \\ &\quad + \log p_{\text{pr}}(\boldsymbol{\theta}) - \log \hat{m}_r(\boldsymbol{\theta}), \end{aligned} \quad (13)$$

$$J_T(\boldsymbol{\theta}) = - \sum_j [\log \theta_j + \log \sigma_j + \log(\ln 10)],$$

where ℓ_c denotes the corner-plot log density, p_{pr} is the adopted prior, $\mathbf{y} = T_\theta(\boldsymbol{\theta})$, and σ_j is the standardization scale of the j th log-parameter coordinate. The J_T term

is the change-of-variables correction from standardized log space to physical parameter space. In the current SAGE figure, p_{pr} is uniform inside the plotted bounds and zero outside them, and the exact finite-pool Top- k selection component of $\hat{m}_r$ is not normalized into a posterior weight. Thus the contours include the available proposal correction needed for physical-coordinate density diagnostics, while remaining explicitly short of calibrated credible regions.

2.4. Candidate Acquisition

At active iteration r , let $\mathcal{D}_r = \{(\boldsymbol{\theta}_i, c_i, E_i, C_i)\}_{i=1}^{N_r}$ denote the current table. After training the proposal model on $\mathcal{D}_r$, the candidate pool is generated as

$$\begin{aligned} \mathbf{y}^{(m)} &\sim q_{\phi_r}(\mathbf{y} \mid c_\star), \quad m = 1, \dots, M_r, \\ \boldsymbol{\theta}^{(m)} &= T_\theta^{-1}(\mathbf{y}^{(m)}). \end{aligned} \quad (14)$$

Here ϕ_r denotes the flow parameters after iteration r , M_r is the raw candidate-pool size, and T_θ^{-1} maps standardized log parameters back to physical parameters. Equation (14) is the direct learned-proposal distribution used for acquisition; the posterior-diagnostic correction in Equation (12) is not applied when choosing the next simulation. These candidates are learned-proposal draws, not a new Latin-hypercube design. Candidate generation enforces parameter bounds, recomputes the post-clipping log density, removes duplicate rounded parameter signatures, and keeps the highest-ranked rows. The large candidate count is a cheap proposal count, not a physical-evaluation count.

Direct learned-proposal acquisition uses the target-conditioned density itself to choose the next evaluation. In the one-shot form, $\boldsymbol{\theta}_{r+1}$ is drawn directly from $q_{\phi_r}(\boldsymbol{\theta} \mid c_\star)$. In the finite-pool form used for the first SAGE comparisons, M_r learned-proposal candidates are drawn and ranked by the learned joint density

$$s_{\text{flow}}(\boldsymbol{\theta}) = \log q_{\phi_r}(T_\theta(\boldsymbol{\theta}) \mid c_\star). \quad (15)$$

The highest-score member is selected, and only that one selected parameter vector is evaluated by SAGE at the active step. This rule is stochastic because the candidate pool is drawn from the proposal density, but it is more exploitative than a single direct draw because it retains the highest-density sample from that pool.

The corresponding MAP-style learned-density selector is more explicitly mode-seeking. This variant starts from many direct learned-proposal samples, optimizes their standardized log-parameter coordinates to increase $\log q_{\phi_r}(T_\theta(\boldsymbol{\theta}) \mid c_\star)$, and then evaluates the highest-density optimized point. The branch is called MAP-style because the optimized target is the learned con-

ditional density over parameters, not a separately evaluated simulator likelihood. Both direct selection and MAP-style selection still evaluate one SAGE model per active iteration; the difference is whether the selected point is a high-scoring proposal draw or an optimized density maximum.

The Top- k or highest-score step is an acquisition rule, not a posterior sampler. It is useful for finding improved calibration points quickly, but it also concentrates later training rows in regions already preferred by the current model. A formal posterior estimator would require either a sequential NPE/APT-style objective with a computable proposal density or a learned likelihood construction followed by a Bayesian sampler (D. S. Greenberg et al. 2019; G. Papamakarios et al. 2019; B. M. Boyd et al. 2026). We therefore evaluate active-learning performance through realized simulation losses, while using corrected density plots only as diagnostics of the learned inverse parameter structure.

This distinction separates direct sampling from MAP-like acquisition. Drawing a single point from q_{ϕ_r} samples the learned calibration density directly. Drawing M_r points and retaining the highest-density member is a stochastic approximation to mode search. Optimizing those candidates gives a still sharper MAP-style selector. This matters in the low-budget regime, because a single learned density can contain seed-specific sharp modes whose forward-model score is not yet known.

A simple mixture with the prior design is therefore useful as a support safeguard, although it is not our default acquisition rule:

$$r_r(\boldsymbol{\theta}) = (1 - \epsilon_r) q_{\phi_r}(\boldsymbol{\theta} \mid c_\star) + \epsilon_r U(\boldsymbol{\theta}), \quad (16)$$

where $U(\boldsymbol{\theta})$ is uniform over the allowed parameter domain and ϵ_r may be decreased over active iterations. This mixture keeps a controlled amount of prior-design support in the acquisition distribution when the adaptive proposal over-concentrates around early training rows. It should be read as an operational guard against proposal-support or prior-design bias, not as a density-ratio posterior correction. For expensive simulations, such a mixture can be combined with a minimum-distance rule,

$$\min_{\boldsymbol{\theta}_i \in \mathcal{D}_r} \left\| \tilde{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_i \right\|_2 > \delta, \quad (17)$$

where tildes denote coordinates normalized to the parameter bounds. This prevents spending an evaluation on an almost duplicated parameter vector. We use these mixture and distance variants as diagnostics for proposal collapse and support coverage; the main SAGE acquisition choice below is between direct learned-proposal selection and MAP-style density maximization.

When seed sensitivity is visible, the MAP-style selector can be replaced by an ensemble-MAP selector. Instead of allowing one independently initialized flow to determine the next evaluation, fit K CRQSF proposal models on the same completed table $\mathcal{D}_r$ and rank candidates by the mixture density

$$\begin{aligned} q_r^{\text{ens}}(\mathbf{y} \mid c_\star) &= \frac{1}{K} \sum_{k=1}^K q_{\phi_r^{(k)}}(\mathbf{y} \mid c_\star), \\ \mathbf{y}_{r+1} &= \arg \max_{\mathbf{y} \in \mathcal{C}_r} q_r^{\text{ens}}(\mathbf{y} \mid c_\star), \\ \boldsymbol{\theta}_{r+1} &= T_\theta^{-1}(\mathbf{y}_{r+1}). \end{aligned} \quad (18)$$

Here $\mathcal{C}_r$ denotes the finite candidate or refined-candidate set. The log score is the density-mixture score $\log \text{sumexp}_k[\log q_{\phi_r^{(k)}}(\mathbf{y} \mid c_\star)] - \log K$, not the mean member log density. The ensemble reduces sensitivity to a seed-specific sharp mode while preserving the MAP-like acquisition logic, but it does not by itself remove adaptive-sampling bias.

The same ensemble density also defines an ensemble direct-sampling branch,

$$\begin{aligned} \mathbf{y}_{r+1} &\sim q_r^{\text{ens}}(\mathbf{y} \mid c_\star), \\ \boldsymbol{\theta}_{r+1} &= T_\theta^{-1}(\mathbf{y}_{r+1}). \end{aligned} \quad (19)$$

This branch averages over independently trained proposal models but retains the one-shot sampling behavior of the direct learned-proposal selector. It is therefore less mode-seeking than ensemble MAP: a seed-specific sharp density peak must affect the mixture distribution before it can change the next draw, but the selected point remains a typical stochastic proposal rather than an optimized density maximum. In implementation, this rule can be run with the same K , training table, target context, parameter bounds, and active evaluation budget as ensemble MAP, changing only the final acquisition step from $\arg \max q_r^{\text{ens}}$ to a direct draw from q_r^{ens} .

In branches that use historical-neighborhood evidence, the base ranking score is

$$s_{\text{base}}(\boldsymbol{\theta}) = \log q_{\phi_r}(T_\theta(\boldsymbol{\theta}) \mid c_\star) + \log w_{\text{hist}}(\boldsymbol{\theta}), \quad (20)$$

where w_{hist} favors candidate regions whose nearby completed simulations produced contexts close to the target. Exploration is supplied by the Latin-hypercube design, stochastic proposal draws, the finite candidate pool, and nearest-neighbor exclusion, rather than by a separate uncertainty-acquisition term. Setting $w_{\text{hist}} = 1$ recovers pure learned-density ranking, and explicit nearest-neighbor exclusion in log-parameter space prevents reselection of already evaluated parameter vectors.

2.5. Cost-Aware Acquisition

For the Sedov refinement-complexity cost variant, the production cost surrogate is a Gaussian-process regressor with an automatic-relevance-determination radial-basis kernel. Let $\tilde{\boldsymbol{\ell}}$ be the standardized log-parameter vector. The kernel is

$$k(\tilde{\boldsymbol{\ell}}, \tilde{\boldsymbol{\ell}}') = \exp \left[-\frac{1}{2} \sum_d \frac{(\tilde{\ell}_d - \tilde{\ell}'_d)^2}{\lambda_d^2} \right] + \sigma_n^2 \delta_{\tilde{\boldsymbol{\ell}}, \tilde{\boldsymbol{\ell}}'}. \quad (21)$$

The Gaussian process predicts $\hat{z}(\boldsymbol{\theta})$, where $z = \max[0, \log(t/t_0)]$, and cost-aware acquisition uses $\hat{C} = \exp(\hat{z})$. In offline leave-one-out diagnostics over pooled unique Sedov refinement-complexity records, the automatic-relevance Gaussian-process surrogate gives mean absolute log error 0.0220, versus 0.0297 for the previous ridge-quadratic surrogate; rank correlations are 0.932 and 0.896, respectively. These diagnostics are local cost-model checks, not final science evidence.

The next simulation or reduction ranks calibration value against predicted cost:

$$\boldsymbol{\theta}_{r+1}^\star = \arg \max_{\boldsymbol{\theta} \in \mathcal{A}_r} [s_{\text{base},r}(\boldsymbol{\theta}) - \beta \log \hat{C}_r(\boldsymbol{\theta})]. \quad (22)$$

Equivalently,

$$s_{\text{cost}}(\boldsymbol{\theta}) = s_{\text{base}}(\boldsymbol{\theta}) - \beta \log \hat{C}(\boldsymbol{\theta}). \quad (23)$$

Cost awareness changes candidate ranking, not the physical target. In the Sedov two-row physical-improvement branch, the learned proposal model generates the finite pool, while final ranking uses a forward profile surrogate or nearby observed metric errors in log-parameter space:

$$\begin{aligned} \widehat{\Delta E}_r(\boldsymbol{\theta}) &= \max [E_r^{\text{best}} - \hat{E}_r(\boldsymbol{\theta}), \epsilon], \\ s_{\text{cost}}(\boldsymbol{\theta}) &= \log \widehat{\Delta E}_r(\boldsymbol{\theta}) - \beta \log \hat{C}_r(\boldsymbol{\theta}). \end{aligned} \quad (24)$$

This branch tests an expected physical-error decrease per refinement-complexity criterion. The stored proposal log density remains diagnostic in this branch and is not added to Equation (24).

2.6. Case-Specific Implementation

The Sedov–Taylor blast wave is a controlled hydrodynamic benchmark with an analytical self-similar solution (L. I. Sedov 1959; G. I. Taylor 1950a,b). The calibration target is not a scalar shock-radius fit alone: each completed simulation is reduced to fixed-radius profile differences relative to the analytical solution,

$$c_{\text{profile}} = \{\Delta \rho(r_b), \Delta P(r_b), \Delta u(r_b), \Delta e(r_b)\}_{b=1}^{N_b}. \quad (25)$$

The three calibrated parameters are Enzo AMR refinement thresholds,

$$\boldsymbol{\theta} = (\theta_{\text{slope}}, \theta_{\text{pressure}}, \theta_{\text{energy}}), \quad \theta_j \in [10^{-2}, 10^3]. \quad (26)$$

Sedov cost-aware branches use refinement complexity as the computational-cost target, so the same profile-calibration problem can be tested with and without an explicit AMR-complexity penalty.

The Rotating Cylinder problem is the second Enzo closed-loop calibration test (M. J. Berger & P. Colella 1989; G. L. Bryan et al. 2014). The calibrated parameters are $\theta_1 = \text{MinimumOverDensityForRefinement}$ and $\theta_2 = \text{MinimumShearForRefinement}$; `MinimumEfficiency` is not calibrated. Each simulation is reduced to a profile tensor $p(\boldsymbol{\theta}) \in \mathbb{R}^{4 \times 12}$:

$$p = (\Delta L_x, \Delta L_y, \Delta L_z, L_z/L_z(0) - 1), \quad p_\star = 0. \quad (27)$$

The scalar physical error is the RMS distance from this zero target, and the runtime-aware branch uses measured wall-clock runtime as the cost target.

The SAGE case extends the active-learning workflow to a semi-analytic model (SAM) of galaxy formation. SAGE is run locally on the mini-Millennium example merger-tree input distributed with the code (D. J. Croton et al. 2016). This is a pipeline-level reproduction setting: it tests parameter patching, metric extraction, observational target construction, and active-learning behavior on the local executable SAGE path, but it is not a numerical reproduction of the full Millennium-volume SAGE calibration.

The five-observable Croton-style baseline calibrates seven scale parameters,

$$s_{\text{SAGE5}} = (s_{\text{SF}}, s_{\text{yield}}, s_{\text{reheat}}, s_{\text{eject}}, s_{\text{radio}}, s_{\text{quasar}}, s_{\text{BH}}), \quad (28)$$

and the eight-observable extension calibrates

$$s_{\text{SAGE8}} = (s_{\text{b}}, s_{\text{SF}}, s_{\text{yield}}, s_{\text{reheat}}, s_{\text{eject}}, s_{\text{radio}}, s_{\text{quasar}}, s_{\text{BH}}). \quad (29)$$

The eight-observable target adds gas fraction, stellar-mass-density evolution, and quenched fraction to the five Croton-style families. The SAGE comparison also evaluates the local manual/default parameter vector in the same metric pipeline, so the active-learning result can be read against both the initial LHS and the local manual baseline.

3. RESULTS

3.1. Sedov–Taylor Calibration

The Sedov AMR thresholds are calibrated because they affect physical profile accuracy, not only runtime. Direct sensitivity tests over the completed LHS100

and LHS10 Sedov runs show that higher threshold values generally increase the visible density–pressure–velocity profile error. The dominant one-dimensional lever is `MinimumSlopeForRefinement`, with visible profile Spearman correlation 0.778. The pressure-jump and energy-ratio thresholds are weaker but still positive directions, with corresponding visible-profile correlations 0.382 and 0.378. Thus, the slope threshold is the most impactful AMR parameter in the Sedov calibration, while the other two thresholds still contribute to the joint calibration surface. The detailed sensitivity plot and tercile comparison are given in Appendix A.

Post-hoc physical quality is summarized from the fixed-grid channel root-mean-square error. The trajectory plots emphasize the visible-profile diagnostic,

$$E_{\text{vis}}(\boldsymbol{\theta}) = \frac{1}{3} (\text{error}_\rho + \text{error}_P + \text{error}_u), \quad (30)$$

because the specific-internal-energy residual is numerically large and nearly constant across completed active rows.

The 100-row initial-design branch starts from 100 log-spaced Latin-hypercube seed simulations, appends 40 active rows in two-sample batches, and uses a 256-bin fixed profile grid. Its best seed row has normalized visible-profile score 0.633; the best active row has score 0.637, and the median active score is 0.663. The 10-row initial-design branch starts from 10 log-spaced Latin-hypercube seed simulations, appends 20 one-at-a-time active rows, and uses the 768-bin split profile grid. Its best seed row has score 0.835; the best active row has score 0.842, and the median active score is 0.858.

Branch	Eval.	Best seed	Best active	Median active
100-row seed	100+40	0.633	0.637	0.663
10-row seed	10+20	0.835	0.842	0.858

Table 1. Sedov initial-design result summary. Scores are normalized visible-profile errors from density, pressure, and velocity; lower is better.

The two-row full-range branch is a small-data stress test rather than a solved calibration run. It starts from two full-range log-spaced Latin-hypercube seed simulations and uses the same 768-bin profile context as the 10-row branch. The full-range run reaches the configured 20 active iterations without meeting the tolerance criterion, but its active rows still reduce the visible-channel normalized profile errors relative to the two-row seed scale: the mean active normalized root-mean-square error is 0.854 for density, 0.788 for pressure, and 0.830 for velocity.

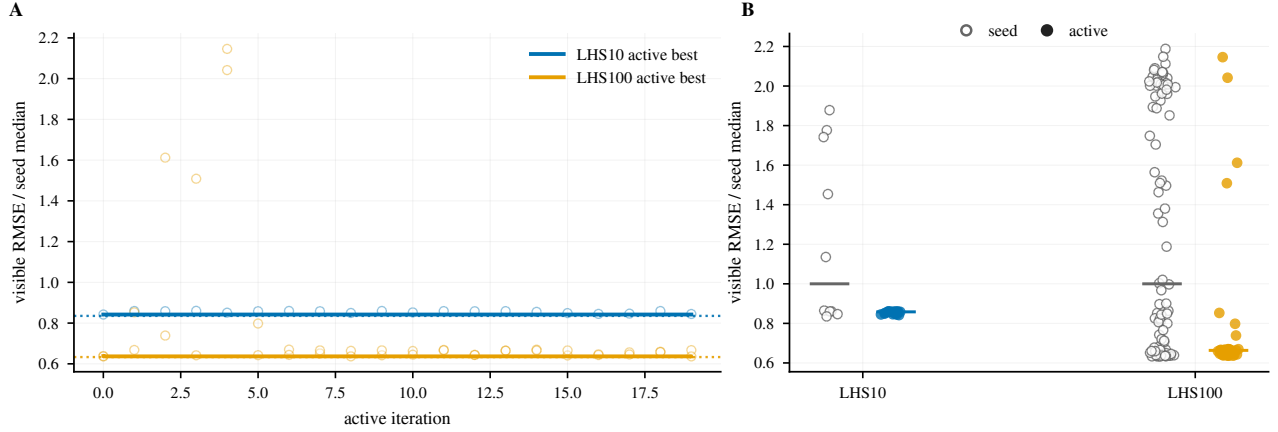


Figure 2. Sedov active-learning comparison for 10-row and 100-row Latin-hypercube-sample seeds. Both branches generate valid active parameter vectors and approach the best seed scale, but neither branch clearly improves over its best initial-design row.

The problematic behavior appears in the refinement profile, not in the physical Sedov shock location. The high-complexity weighted branch has an additional inner refinement ring near $r \simeq 0.06$, while the outer ring remains the main blast shock near $r \simeq 0.10$. Controlled ablations show that this inner ring is triggered by the lower slope threshold. Thus the two-row two-peak issue is a cost failure mode: physical-error diagnostics do not show a second shock, but the selected adaptive-mesh refinement pattern materially increases mesh complexity.

The two-peak diagnosis motivates a cost-aware acquisition rule for Sedov. The new 10-row cost-aware branch uses the learned proposal model to generate a finite target-conditioned candidate pool, nearest-neighbor evidence to estimate expected physical-error decrease from nearby observed rows, an automatic-relevance Gaussian-process model to predict refinement-complexity cost, and score-resampling selection with $\tau_{\text{acq}} = 0.1$. The implemented score is

$$s_{\text{cost}}(\theta) = \log \widehat{\Delta E}_{\text{near}}(\theta) - \beta \log \widehat{C}_{\text{cost}}(\theta). \quad (31)$$

The first completed cost-aware 10-row proposal has visible smooth profile error 0.0456, compared with 0.0506 for the best initial-design row under the same diagnostic. Its measured refinement complexity is 113,446, while the Gaussian-process surrogate predicted 100,589 (actual/predicted = 1.13). Most importantly, the radial refinement profile has a single mean-level peak at $r = 0.1075$, aligned with the simulated shock radius $R_s = 0.1062$ and the analytical shock radius $R_s = 0.1072$.

3.2. Rotating Cylinder Calibration

The Rotating Cylinder profile-conditioned proposal model uses the Sedov modeling idea with the angular-

momentum tensor in Equation (27). The target-only baseline uses s_{base} without a runtime penalty. It isolates target-conditioned proposal quality when every selected point launches a new Enzo simulation. The branch tests whether the shared profile-conditioned interface can move from a Sedov analytical residual tensor to a time-dependent angular-momentum conservation target.

The cost-aware Rotating Cylinder branch uses measured wall-clock runtime and a ridge-quadratic runtime surrogate. Its runtime-aware acquisition score is

$$s_{\text{RC,cost}}(\theta) = s_{\text{base}}(\theta) - 0.5 \log \widehat{C}(\theta). \quad (32)$$

The target-only branch uses 10 initial rows plus 4 active rows, while the runtime-aware branch uses 10 initial rows plus 10 active rows. Because the branches are not replicated under matched budgets, physical error, runtime, cost-aware rank, marginal efficiency, and Pareto status must be interpreted together.

3.3. SAGE Calibration

The SAGE case changes the evidence surface from hydrodynamic profile tensors to an observational metric vector evaluated by a semi-analytic model on merger trees. The eight-observable target adds gas fraction, stellar-mass-density evolution, and quenched fraction to the five Croton-style families, testing baryon-cycle and quenching constraints that the classical diagnostics do not fully fix. Each family receives equal weight. Missing target bins are imputed to the target value plus five sigma before scoring, penalizing absent model output without making the objective undefined.

The eight-observable target starts from the same 16-row initial Latin-hypercube design and uses 10 active SAGE evaluations per branch. No nearest-neighbor

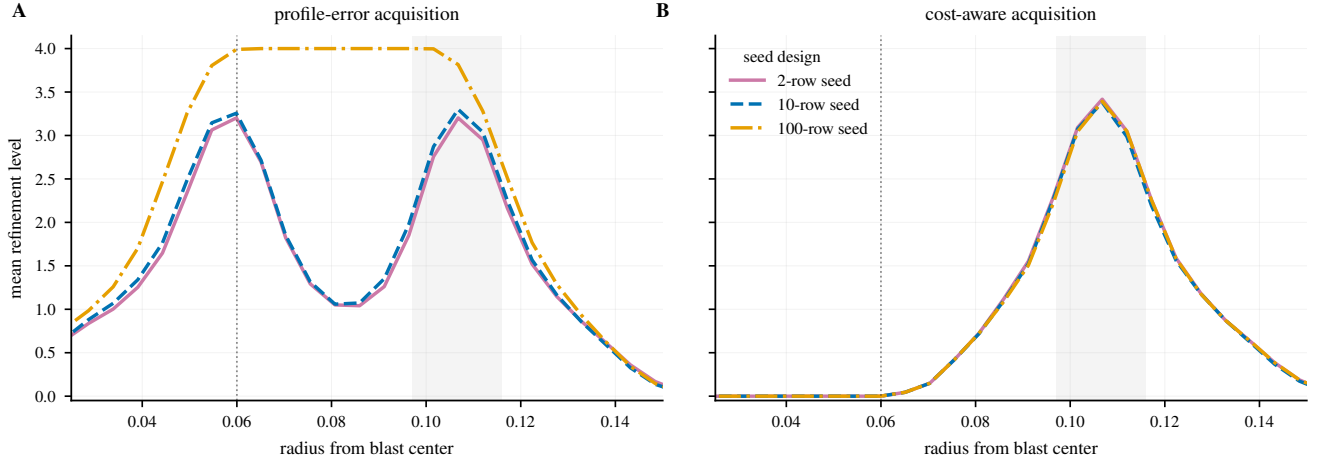


Figure 3. Sedov radial mean refinement profiles before and after introducing the cost-aware acquisition rule. Under profile-error acquisition alone, the representative two-row branch develops an inner adaptive-mesh refinement peak near $r \simeq 0.06$ in addition to the shock-centered peak. With cost-aware acquisition, the representative accepted rows place the dominant refinement peak at the Sedov shock and suppress the inner two-row refinement ring.

observational-score reranking and no cost-aware penalty are used in these SAGE branches. The acquisition comparison is intentionally restricted to direct learned-proposal selection and MAP-style density maximization. The shared best initial-design row has $R^2 = 0.4359$. A single-flow direct sample-and-rank selector reaches $R^2 = 0.7060$, a gain of $\Delta R^2 = 0.2701$, while the matched single-flow MAP-style selector reaches $R^2 = 0.6302$ and ends at $R^2 = 0.2883$. This initial comparison shows that a single learned-density maximum can be a poor final active point even when the learned proposal itself is informative.

The random-seed tests change the methodological conclusion from “avoid MAP” to “average before applying MAP.” Across five CRQSF random seeds, direct sampling can find high-scoring points but has large iteration-to-iteration and seed-to-seed variability. The ensemble-MAP branch therefore fits five CRQSF members with seeds 41–45 and selects the maximum of their mixture density (Equation (18)). It reaches $R^2 = 0.6851$, only below the best single direct-sample seed but above the direct-sample mean, and it retains the MAP-like exploitation logic while reducing sensitivity to a seed-specific sharp mode. The matched ensemble direct-sampling branch reaches only $R^2 = 0.5060$ and finishes at $R^2 = 0.3460$, showing that ensembling alone is insufficient if the acquisition step remains a one-shot stochastic draw. Table 2 and Figure 5 summarize the direct-versus-MAP selector comparison.

Simple-mixture and minimum-distance variants are retained as support-bias diagnostics rather than as the main acquisition family. In the seed sweep, the simple-mixture plus minimum-distance branch has one of the

better mean final scores, $R^2 = 0.499 \pm 0.163$, but a larger best-active spread than the direct/MAP selectors. We therefore interpret it as a guard against proposal-collapse or prior-design support bias, not as the default replacement for ensemble MAP.

Selector	Initial	Active	ΔR^2	Final
Direct sample-and-rank	0.4359	0.7060	0.2701	0.4859
Single-flow MAP-style	0.4359	0.6302	0.1943	0.2883
Ensemble MAP, $K = 5$	0.4359	0.6851	0.2491	0.4653
Ensemble direct sample, $K = 5$	0.4359	0.5060	0.0700	0.3460

Table 2. Eight-observable SAGE calibration summary for the direct-sampling and MAP-family learned-proposal selector comparison. The initial Latin-hypercube design, scale range $[0.3, 5]$, target definition, and 10-active-evaluation budget are held fixed. Higher R^2 is better. Direct sample-and-rank is the strong single-flow baseline; ensemble MAP is the recommended acquisition rule after accounting for CRQSF random-seed sensitivity.

Table 3 lists the best active scale vector. Figures 6 and 7 summarize this single-flow direct sample-and-rank result across the observable families and aggregate metrics.

The conditional density learned at the final trained proposal state is sampled with random-walk Metropolis chains in standardized log-parameter space after a direct-flow prescan. The chain target uses the physical-coordinate correction in Equation (13) and enforces the SAGE parameter bounds; the exact $\hat{m}_r$ term induced by deterministic Top- k selection is not treated as a calibrated probability. One- and two-dimensional density estimates are kernel-smoothed for visualization (Fig-

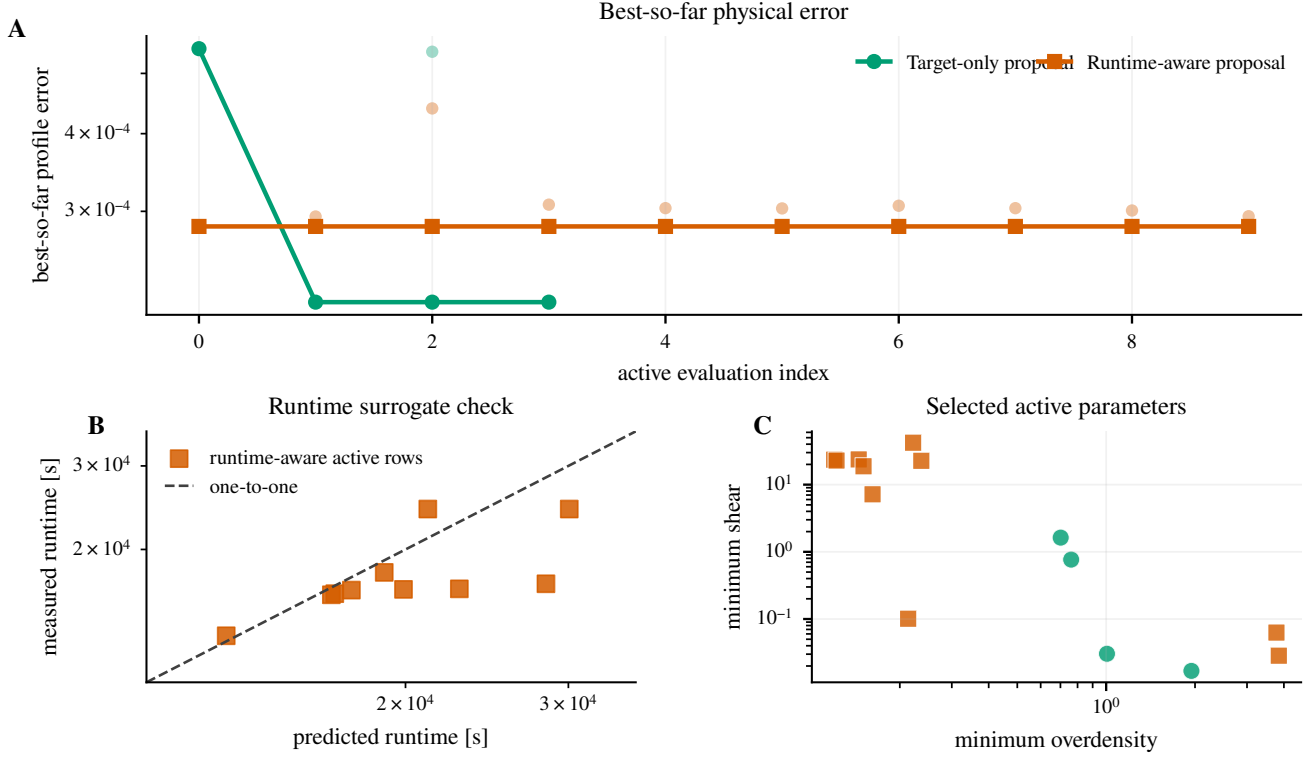


Figure 4. Rotating Cylinder descriptive comparison for pure and runtime-aware learned-proposal acquisition. The panels summarize best-so-far profile root-mean-square error, predicted versus measured runtime for the runtime-aware branch, and selected active points in the two-dimensional Enzo refinement-threshold space.

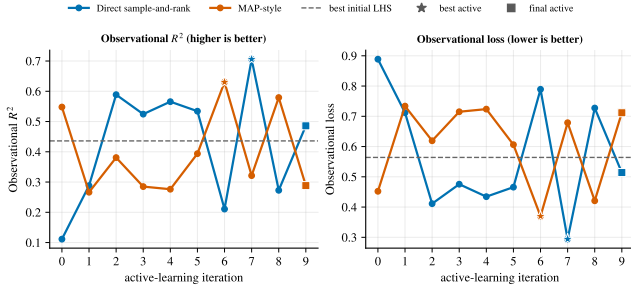


Figure 5. Eight-observable SAGE comparison between direct joint sample-and-rank acquisition and the MAP-style learned-density ablation. Stars mark the best active row and squares mark the final active row. Both branches evaluate one SAGE parameter vector per active iteration; the difference is whether the selected point is the highest-ranked direct sample or the highest-density optimized point.

ure 8). The density is concentrated away from the fiducial scale vector for several feedback-related parameters and shows non-Gaussian, multi-modal parameter correlations. We interpret this plot as a proposal-bias-aware diagnostic of the learned inverse parameter structure, rather than as a fully calibrated Bayesian posterior over the physical SAGE parameters. Table 4 reports the corresponding peak-level forward evaluations.

Scale parameter	Best active value
s_{baryon}	2.2014
$s_{\text{star formation}}$	0.8130
s_{yield}	3.7622
s_{reheat}	4.7175
s_{eject}	5.0000
s_{radio}	4.5319
s_{quasar}	4.3277
$s_{\text{black hole}}$	2.9211

Table 3. Best active eight-parameter scale vector from the single-flow direct sample-and-rank branch. All values are multiplicative factors relative to the SAGE paper/default parameter vector.

4. DISCUSSION

4.1. Sedov: Resource Management in AMR Calibrations

The Sedov experiments show that calibration in adaptive-mesh simulations is not only a question of reducing physical residuals. A profile-error-only acquisition rule can allocate computational resources inefficiently because it does not distinguish between parameter choices that improve the target profile with moderate mesh complexity and choices that achieve similar profile scores by triggering much more refinement. This mat-

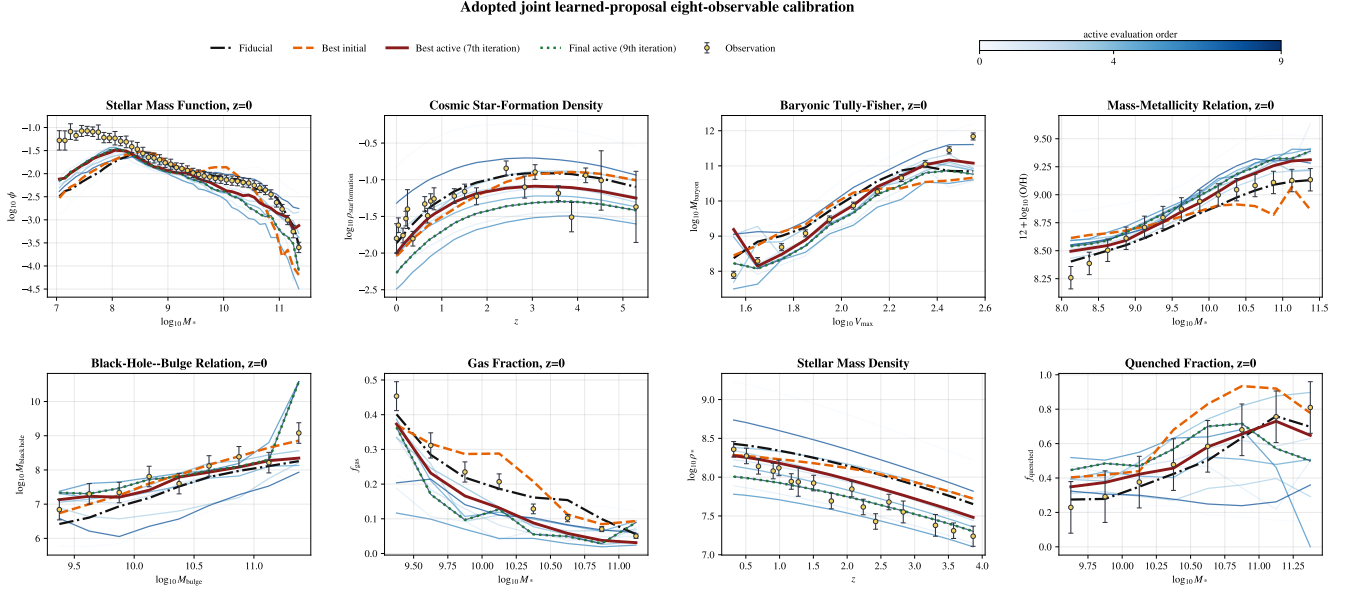


Figure 6. Eight-observable SAGE active-learning overview for the single-flow direct sample-and-rank branch. The panels compare the fiducial SAGE row, shared best initial row, best active-learning row, final active-learning row, and observational constraints across the eight target families. Blue curves show the active proposal sequence, with darker colors indicating later evaluations.

Peak	Fraction	R^2	Loss	Missing
1	0.424	0.481	0.519	1
2	0.174	0.572	0.428	0
3	0.210	0.505	0.495	0
4	0.192	0.447	0.553	1

Table 4. Four-peak diagnostic for the single-flow direct-rank SAGE conditional-density sample. Peaks are separated in normalized eight-parameter space and assigned retained Markov-chain samples. The R^2 and loss columns report new SAGE forward evaluations at the maximum-density sample within each peak, using the same observational metric extraction and scoring pipeline as the active-learning runs. Missing bins denote target observational bins absent from the model extraction and imputed with the existing missing-metric penalty; lower loss and higher R^2 indicate better target agreement.

ters for AMR calibration because the refinement pattern is part of the realized simulation cost, not just an implementation detail behind the physical metric.

Including refinement complexity in the acquisition score adds information that is not captured by density, pressure, or velocity profile errors alone. The cost-aware Sedov branch therefore reframes the selection problem as a trade-off between physical accuracy and AMR complexity. In this form, the calibration loop can prefer candidates that preserve acceptable blast-wave agreement while avoiding unnecessarily expensive refinement behavior. The broader implication is that cost-aware calibration is especially important for compu-

tationally expensive AMR simulations, where repeated active-learning evaluations can otherwise spend limited resources on numerically plausible but inefficient mesh configurations.

4.2. SAGE

The SAGE experiment demonstrates that the same learned-proposal machinery can improve a multi-observable semi-analytic-model calibration with only ten additional model evaluations. Rather than relying on an unguided extension of the initial design, the joint learned-proposal selector explores parameter combinations that are explicitly conditioned on the observational target vector. This target-directed exploration is useful in the eight-observable setting because the observables constrain overlapping but non-identical parts of the parameter space.

The proposal-strategy tests clarify that the central acquisition choice is between direct sampling from the learned proposal and MAP-style optimization of that proposal. Both families run only one SAGE model at each active iteration, so the distinction is not the number of physical evaluations. The distinction is how that single point is obtained. Direct sample-and-rank selects the best member of a finite stochastic pool drawn from the joint learned proposal. It can find excellent active points, including the highest single-flow result in Table 2, but it also has substantial iteration-to-iteration variability and can end away from the best discovered point.

Adopted joint learned-proposal evaluation metrics

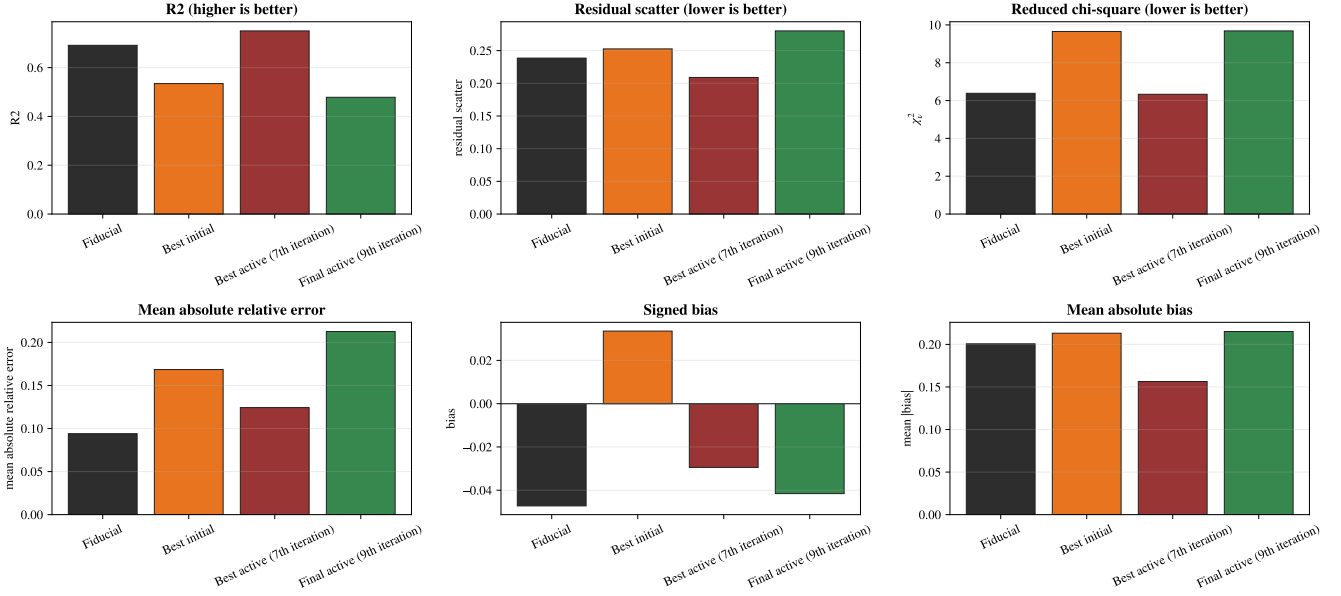


Figure 7. Aggregate evaluation metrics for the single-flow direct sample-and-rank eight-observable SAGE branch. The bars compare the fiducial row, the shared best initial row, the best active-learning row, and the final active-learning row using weighted R^2 , root-mean-square residual, reduced χ^2 , mean absolute relative error, signed bias, and mean absolute bias.

Single-flow MAP-style selection is the opposite failure mode. It pushes many sampled starting points toward local maxima of the learned density and then selects the highest optimized point. This is attractive when the learned proposal is accurate, but the SAGE tests show that a single independently trained CRQSF model can contain a sharp seed-specific mode whose forward observational score is not optimal. The ensemble-MAP selector is therefore the more natural acquisition rule: it preserves mode-seeking exploitation while requiring the selected maximum to be supported by the mixture of several independently trained proposal models. The ensemble direct-sampling branch supports this interpretation. Averaging over proposal seeds helps define a better acquisition density, but a one-shot draw from that density remains a typical stochastic proposal rather than a high-density calibration candidate.

Simple-mixture and minimum-distance variants address a different issue. They are useful when the adaptive proposal over-concentrates and stops covering the prior-supported parameter volume, or when consecutive active evaluations become nearly duplicated. Their role is therefore to guard against proposal-support or prior-design bias, not to provide a fully corrected posterior density and not to replace ensemble MAP as the default acquisition rule.

This distinction also sets the success criterion for the SAGE experiment. The active-calibration claim is based

on the best realized forward-model score under a fixed evaluation budget, not on convergence of the learned density itself. A corner plot may continue to change with only 26 total SAGE evaluations, while the best observed calibration row has already improved. The reported recommendation is therefore the best evaluated SAGE parameter vector, not automatically the final active sample or the highest-density point of the learned proposal.

The best SAGE solution favors increased values for several stellar-feedback and AGN-feedback scale parameters relative to the fiducial vector. This pattern is astrophysically interpretable as a preference for stronger regulation of star formation, gas retention, and galaxy growth in the local mini-Millennium calibration. At the same time, the learned density remains multi-modal and correlated, indicating parameter degeneracies and residual observational tensions rather than a single uniquely determined physical solution. The SAGE result should therefore be read as a target-directed calibration improvement and parameter-interpretation diagnostic, not as a final full-volume recalibration of the model.

4.3. Transferability of the Pipeline

Across the three cases, the main methodological value is the separation between case-specific interfaces and a reusable calibration engine. Sedov, Rotating Cylinder, and SAGE require different executable wrappers, target representations, and metric extractors, but each case

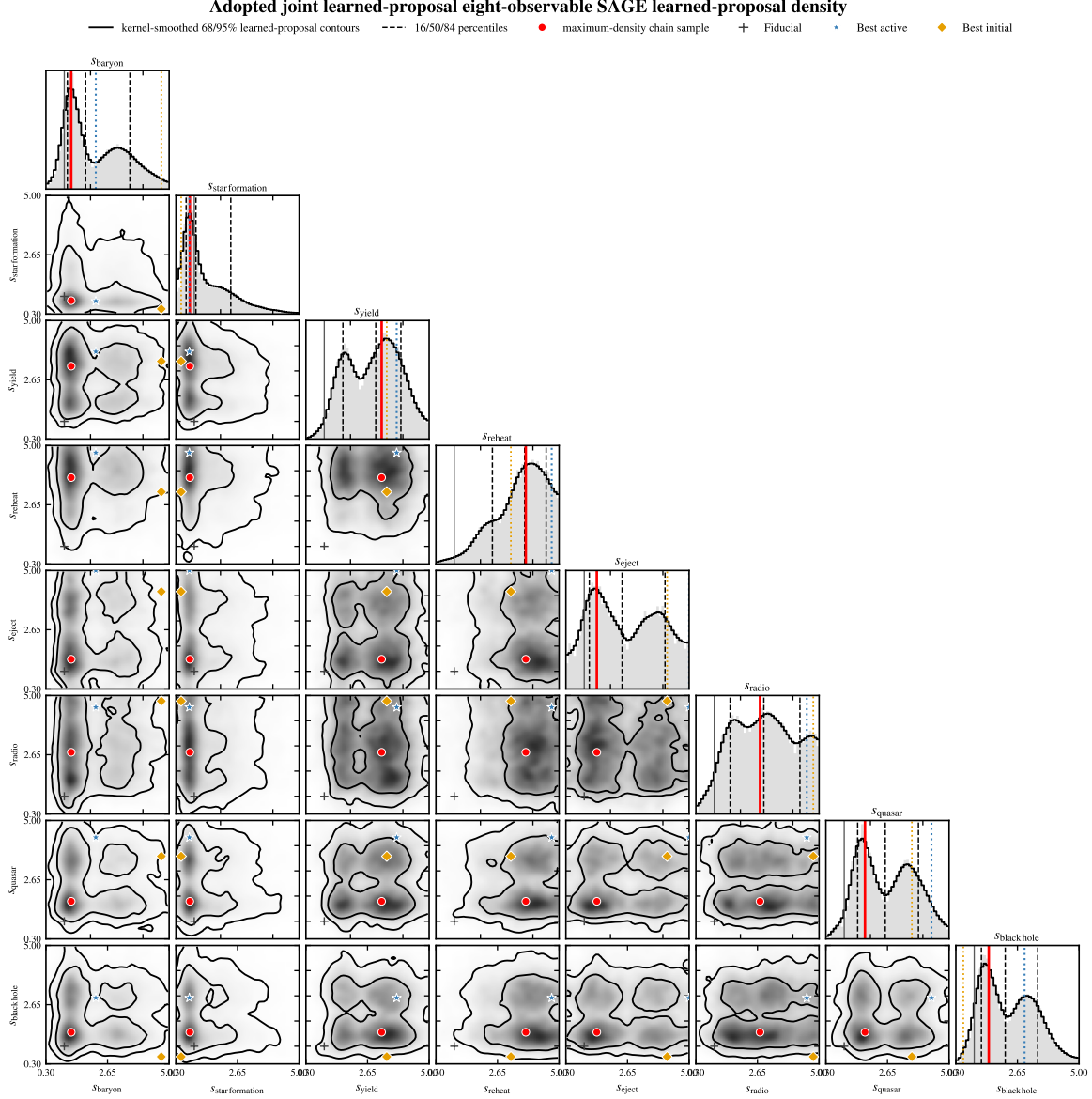


Figure 8. Corrected learned-density corner plot for the single-flow direct sample-and-rank eight-observable SAGE branch. Diagonal panels show kernel-smoothed one-dimensional marginal densities with 16th, 50th, and 84th percentile markers; off-diagonal panels show kernel-smoothed 68 and 95 percent density contours from retained Markov-chain samples. The red marker denotes the sample with maximum physical-space density, the blue star denotes the best evaluated active-learning row, the orange diamond denotes the best initial row, and the gray plus denotes the fiducial SAGE parameter vector. The plot exposes nonlinear degeneracies among the calibrated SAGE scale parameters after the physical-parameter density correction, but should be read as a proposal-bias-aware diagnostic rather than a fully calibrated posterior.

produces the same kind of training table for the learned proposal model. This shared structure allows the proposal model and acquisition logic to remain fixed while the target context changes from an analytical hydrodynamic profile to a conservation history and then to a multi-observable galaxy-formation metric vector.

The benefit of target-conditioned parameter proposals is that the active learning step is driven by the desired target rather than by a single hand-coded simulator-

specific rule. Case-specific interfaces still matter: they define the physical target, enforce parameter transformations, execute the model, and extract the metrics. The reusable component is the outer calibration loop that learns from completed rows, proposes target-conditioned parameters, and selects new evaluations. This division is what makes the pipeline portable without pretending that the physics reductions are identical across cases.

4.4. Limitations

Several limitations follow from this scope. The evaluation budgets are small, and stochastic replicates are under-sampled, especially for the hydrodynamic branches where runtime and refinement behavior can change across nearby parameter settings. The calibration result also depends on the chosen target metrics and observable weighting. In SAGE, changing the observable configuration changes the learned parameter structure, so the reported parameter interpretation is conditional on the adopted observable families, equal-family weighting, missing-bin treatment, and local reduction path. The low-budget density estimates should also be interpreted separately from the calibration recommendation: a stable or improving J_{best} curve is stronger evidence for successful active calibration than visual corner-plot convergence.

The current cost-surrogate evidence is strongest for AMR refinement complexity. Extending cost awareness beyond AMR simulations will require case-specific cost features and surrogate validation. The SAGE results are also limited to the local mini-Millennium setting; full-volume evaluation is needed before drawing stronger astrophysical conclusions. A SAGE-specific cost-aware branch remains future work because runtime, merger-tree size, post-processing effort, and metric-extraction cost would need to be recorded consistently before defining a defensible SAM cost target.

Finally, the conditional density is used directly as a target-conditioned proposal model during active learning. The corner plots apply the available physical-density correction and explicitly account for the proposal-bias limitation when interpreting learned parameter structure, degeneracy, and multi-modality. Deterministic Top- k selection prevents us from treating those diagnostics as fully calibrated posterior probability statements. Exact proposal correction, density-ratio correction, or a dedicated simulation-based-inference posterior objective would be needed before interpreting the learned density as a calibrated posterior over physical parameters with uncertainty quantification. Future active-calibration variants should treat ensemble MAP as the default SAGE acquisition rule, with one-shot direct sampling retained as a stochastic baseline and simple-mixture/minimum-distance guards tested only when support coverage or proposal collapse is the specific concern.

5. SUMMARY

Astrophysical calibration requires parameter choices that improve agreement with physical or observational targets under limited evaluation budgets. We present a

reusable machine-guided calibration pipeline that separates case-specific target construction, execution, and metric extraction from a fixed automated engine based on CRQSF proposal learning, finite-pool candidate selection, and optional cost-aware reranking. We applied the same structure to three cases: Sedov–Taylor profile calibration, Rotating Cylinder angular-momentum calibration, and local SAGE semi-analytic-model calibration.

We summarize our main results as follows:

1. **Pipeline transferability.** The same target representation $\rightarrow$ proposal model $\rightarrow$ candidate acquisition $\rightarrow$ evaluation loop is used across an analytical hydrodynamic benchmark, a second Enzo conservation target, and a semi-analytic galaxy-formation model.
2. **Sedov profile calibration.** The Sedov AMR thresholds affect visible profile accuracy. `MinimumSlopeForRefinement` is the most impactful one-dimensional parameter, while the pressure-jump and energy-ratio thresholds remain weaker but positive sensitivity directions.
3. **Sedov cost-aware calibration.** Profile-error-only acquisition can select a parameter basin with excessive inner AMR refinement. Adding the Gaussian-process refinement-complexity surrogate to the acquisition score suppresses that extra refinement peak while preserving the physical profile target.
4. **Rotating Cylinder calibration.** The profile-conditioned interface transfers from Sedov residual profiles to time-dependent angular-momentum conservation profiles, with pure learned-proposal and runtime-aware branches providing a second closed-loop Enzo test.
5. **SAGE calibration.** The eight-observable SAGE branch starts from a shared 16-row initial design. A single-flow direct sample-and-rank branch reaches $R^2 = 0.7060$, while the ensemble-MAP branch reaches $R^2 = 0.6851$ and is the recommended acquisition rule after accounting for CRQSF random-seed sensitivity. The matched ensemble direct-sampling branch reaches only $R^2 = 0.5060$, showing that ensembling is most useful when combined with MAP-style density optimization rather than one-shot sampling.
6. **SAGE parameter interpretation.** The learned SAGE proposal density is concentrated away from

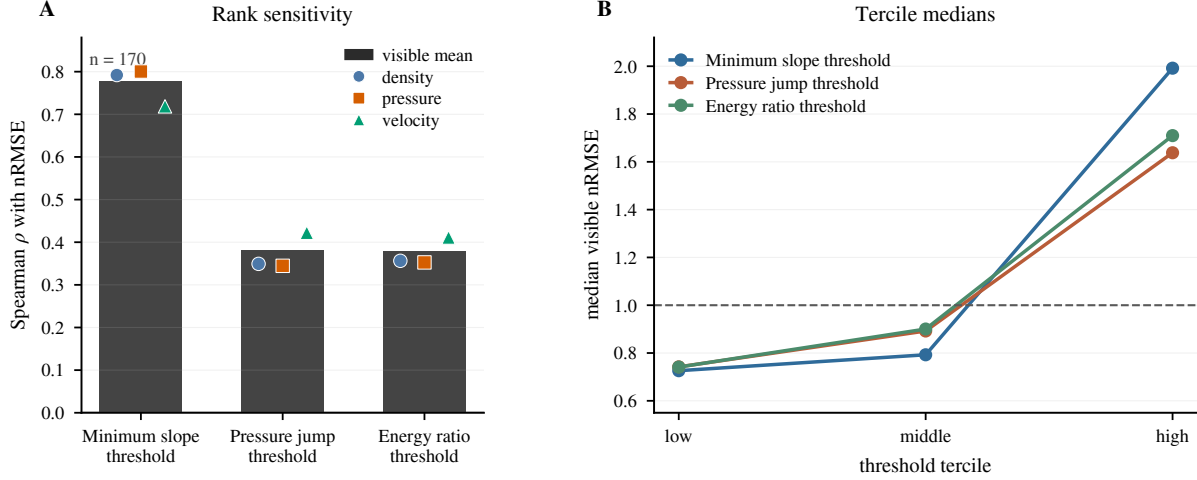


Figure 9. Sedov AMR-threshold sensitivity summary. **(a)** Spearman rank correlations between each log-threshold value and seed-normalized profile nRMSE across 170 completed simulations. Bars show the visible density–pressure–velocity mean; points show the corresponding channel-level correlations. **(b)** Median visible profile nRMSE in the low, middle, and high terciles of each threshold. The high-threshold terciles produce the largest profile residuals, with the slope threshold giving the strongest one-dimensional sensitivity.

the fiducial scale vector for several feedback-related parameters and separates into multiple modes. We interpret these modes as candidate families and proposal-density diagnostics, not as confirmed full-volume astrophysical solutions.

7. Scope and limitations. The evidence supports a portable calibration workflow rather than final astrophysical recalibration. The hydrodynamic branches have limited matched-budget replication, and the SAGE results use local mini-Millennium

runs, fixed observable weights, and the current missing-bin treatment.

8. Future work. The next extensions are full-volume SAGE evaluation, SAGE-specific cost-aware acquisition, systematic tests of target weights and observable definitions, ensemble-MAP replication under broader random-seed and LHS-design sweeps, and explicit posterior correction or uncertainty quantification beyond the present proposal-density diagnostic.

APPENDIX

A. SEDOV PROFILE-ERROR SENSITIVITY TEST

To check whether the three Sedov AMR thresholds need calibration as physical profile controls, we re-analyzed the completed LHS100 and LHS10 Sedov runs as a direct profile-error sensitivity study. The analysis uses 170 completed simulations: 100 LHS100 seed rows, 40 LHS100 active rows, 10 LHS10 seed rows, and 20 LHS10 active rows. For each branch and channel, the radial residual RMSE is normalized by the median seed RMSE in that branch, so the 256-bin LHS100 and 768-bin LHS10 profile grids can be compared on the same relative scale.

Figure 9 shows that all three AMR thresholds affect the physical profile error, with `MinimumSlopeForRefinement` the strongest single-parameter lever. The visible-profile Spearman correlations are 0.778 for the slope threshold, 0.382 for the pressure-jump threshold, and 0.378 for the energy-ratio threshold. For the slope threshold, the low, middle, and high terciles have median visible-profile nRMSE values of 0.726, 0.793, and 1.992, respectively. The same pattern is weaker but still present for the pressure-jump threshold (0.742, 0.892, and 1.638) and the energy-ratio threshold (0.741, 0.900, and 1.710). Thus, large threshold values tend to under-resolve the Sedov shell and increase density, pressure, and velocity residuals. The trend is not a complete one-parameter rule, however: scatter at fixed threshold shows that the three refinement criteria interact, motivating joint calibration with a multivariate active-learning model.

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